



WORK SHEET - 32

SEQUENCE AND PROGRESSION

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.7 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 If $\sum_{r=1}^n T_r = n(n^2 - 1)$ then $\sum_{n=2}^{\infty} \frac{1}{T_n}$ is equal to
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) 1
- Q.2 Let $a_1, a_2, a_3, \dots, a_n$ be a sequence such that $a_1 = 2$ and $a_n - a_{n-1} = n^2 \forall n \geq 2$. If $a_k = 205$ then k is equal to
(A) 7 (B) 8 (C) 11 (D) 15
- Q.3 If a, b, c and d are distinct positive numbers which are in H.P. then the equation $x^2 - kx + (b + c - \sqrt{ac} - \sqrt{bd}) = 0, k \in \mathbb{R}$ has
(A) both roots imaginary (B) both roots real and positive
(C) both roots equal (D) one positive and one negative real root
- Q.4 If the equation $x - \sqrt{x} (a^2 - 3a + 2) + a^2 - 3a + 5 = 0$ has real roots greater than 1 then true set of real values of 'a' is
(A) (1, 2) (B) $(-\infty, 1) \cup (2, \infty)$ (C) $(-\infty, -1] \cup [4, \infty)$ (D) $[-1, 4]$
- Q.5 If sum of first n terms of an A.P. is $S_n = 3n^2 - 2n$, then the value of $\sum_{n=1}^{\infty} \left(\frac{21}{(S_n S_{n+2} + S_{n-1} \cdot S_{n+1}) - (S_n S_{n+1} + S_{n-1} S_{n+2})} \right)$ is equal to
(A) 2 (B) 12 (C) 21 (D) 36
- Q.6 Let α and $\frac{1}{\beta}$ are respectively the first term and common ratio of an infinite G.P. where α and β are the largest and the smallest integral values of x in the domain of the function $f(x) = \ln \sqrt{\sin^{-1} \left(\frac{x-1}{8} \right)}$ for which $g(x) = \frac{5x^2 - 10x + 8}{x - 2}$ is an integer. The sum of the G.P. is equal to
(A) 6 (B) 9 (C) 12 (D) 15
- Q.7 The first term of an A.P. is 1 and the sum of first nine terms equal to 369. The first and the ninth term of a G.P. coincide with the first and the ninth term of the A.P. If the seventh term of the G.P. is T_7 then the value of $\frac{T_7}{3}$ is equal to
(A) 27 (B) 81 (C) 9 (D) 36



[PARAGRAPH TYPE]

Q.8 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question no. 8 to 10

Let $x, \frac{[x]}{2}$ and $\{x\}$, ($x > 0$) are the first three terms of a G.P.

[Note : $[k]$ and $\{k\}$ denote greatest integer and fractional part functions of k respectively.]

Q.8 Common ratio of the G.P. is

- (A) $\tan\left(\frac{\pi}{12}\right)$ (B) $\tan\left(\frac{\pi}{6}\right)$ (C) $\tan\left(\frac{\pi}{8}\right)$ (D) $\tan\left(\frac{3\pi}{8}\right)$

Q.9 If $1 < x < 2$ then product of first three term of the G.P. is

- (A) $\frac{1}{8}$ (B) $(\sqrt{2}-1)^3$ (C) $\frac{1}{4}(\sqrt{2}-1)^2$ (D) $\frac{1}{4}$

Q.10 If $2 < x < 3$ then 20 term of the G.P. is

- (A) $\left(\tan\frac{\pi}{8}\right)^{18}$ (B) $\left(\tan\frac{\pi}{8}\right)^{19}$ (C) $\left(\tan\frac{3\pi}{5}\right)^{19}$ (D) $\left(\tan\frac{3\pi}{5}\right)^{18}$

[MULTIPLE CORRECT CHOICE TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.11 Let $a_1, a_2, a_3, \dots, a_{11}$ be in A.P. and $h_1, h_2, \dots, h_{11}$ be in H.P. If $a_1 = h_1 = 4$ and $a_{11} = h_{11} = 9$ then

- (A) $a_4 h_8 = 36$ (B) $a_4 h_8 = 13$ (C) $\sqrt{a_6 h_6} = 6$ (D) $a_6 + h_6 > 12$

Q.12 If $x, y, z > 0$ and $x + y + z = 9$ then identify the correct statement(s) -

- (A) Maximum value of xyz is 27 (B) Maximum value of xy^2z is $3^8 \cdot 2^{-6}$
(C) Maximum value of x^2yz^3 is $3^8 \cdot 2^{-4}$ (D) Maximum value of $x^2y^3z^4$ is $2^{10} \cdot 3^3$

Q.13 If $x, y, 6, z, 9x$ are five number such that first three are in A.P. last three are in H.P. and 6 is the geometric mean of y and z then three digit number zyx is divisible by

- (A) 2 (B) 3
(C) 157 (D) exactly three prime numbers.

Q.14 Let $M = \underbrace{999\dots9}_{9 \text{ times}}$, $N = \underbrace{999\dots9}_{6 \text{ times}}$ and $P = 999$ then

- (A) $MP = N^2$ (B) $\frac{N}{P} = 1001$
(C) $N - 2P$ is perfect square of a integer (D) $N - P^2$ is an an even integer

[INTEGER TYPE]

Q.1 to Q.3 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.1 Let $a_1, a_2, a_3, \dots$ be in A.P. and $a_1 = 1, a_{102} = 405$. If $\left| a_1^2 - a_2^2 + a_3^2 - a_4^2 + \dots + a_{101}^2 - a_{102}^2 \right| = \prod_{i=1}^5 p_i^{b_i}$

where p_i are prime number and $b_i \in \mathbb{N} \forall i = 1, 2, \dots, 5$, then find the value of $\sum_{i=1}^5 p_i + \sum_{i=1}^5 b_i$.

Q.2 Let a, b, c, d be four distinct natural number in which first three are in HP and last three are in G.P. If

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1 \text{ then find the sum } (a + b + c + d).$$

Q.3 Let T_r be the r^{th} term of an A.P. whose first term is $-\frac{1}{2}$ and common difference is 1, then find the

$$\text{value of } \sum_{r=2}^{101} \left\{ \sqrt{1 + T_r \cdot T_{r+1} \cdot T_{r+2} \cdot T_{r+3}} \right\}.$$

[Note : $\{y\}$ denotes the fraction part function of y .]

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ANSWER KEY

Q.1 B Q.2 B Q.3 D Q.4 C Q.5 A Q.6 B Q.7 C
Q.8 C Q.9 A Q.10 A Q.11 ACD Q.12 ABD Q.13 ABCD Q.14 BCD
Q.15 0065 Q.16 0023 Q.17 0075

